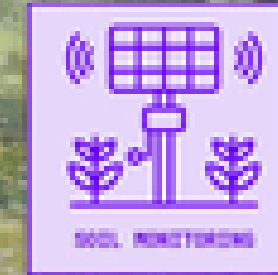
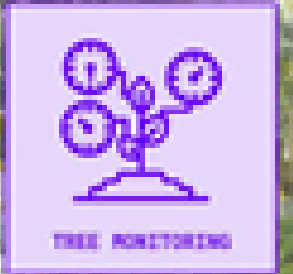
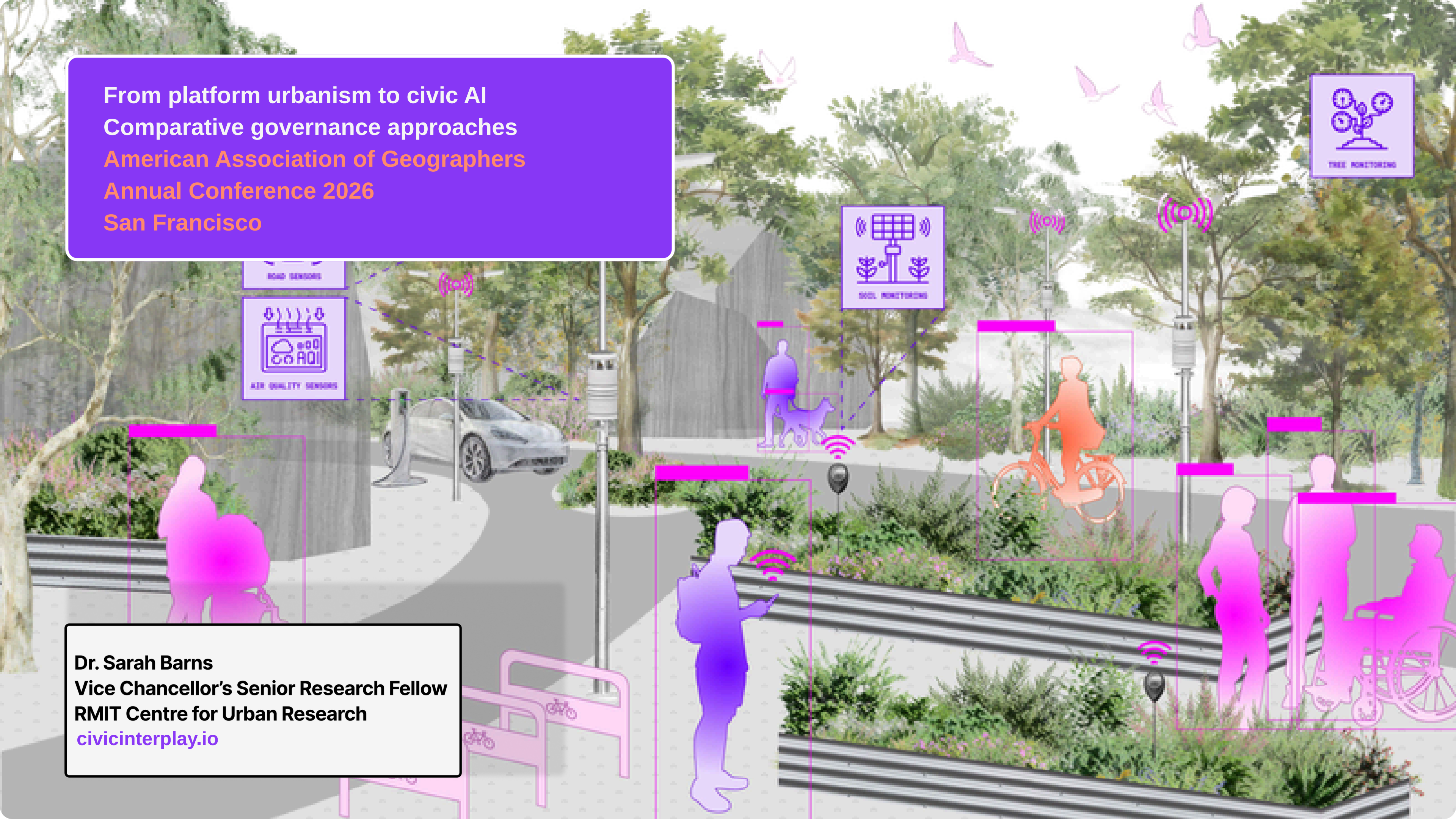


From platform urbanism to civic AI
Comparative governance approaches
American Association of Geographers
Annual Conference 2026
San Francisco



Dr. Sarah Barns
Vice Chancellor's Senior Research Fellow
RMIT Centre for Urban Research
civicinterplay.io

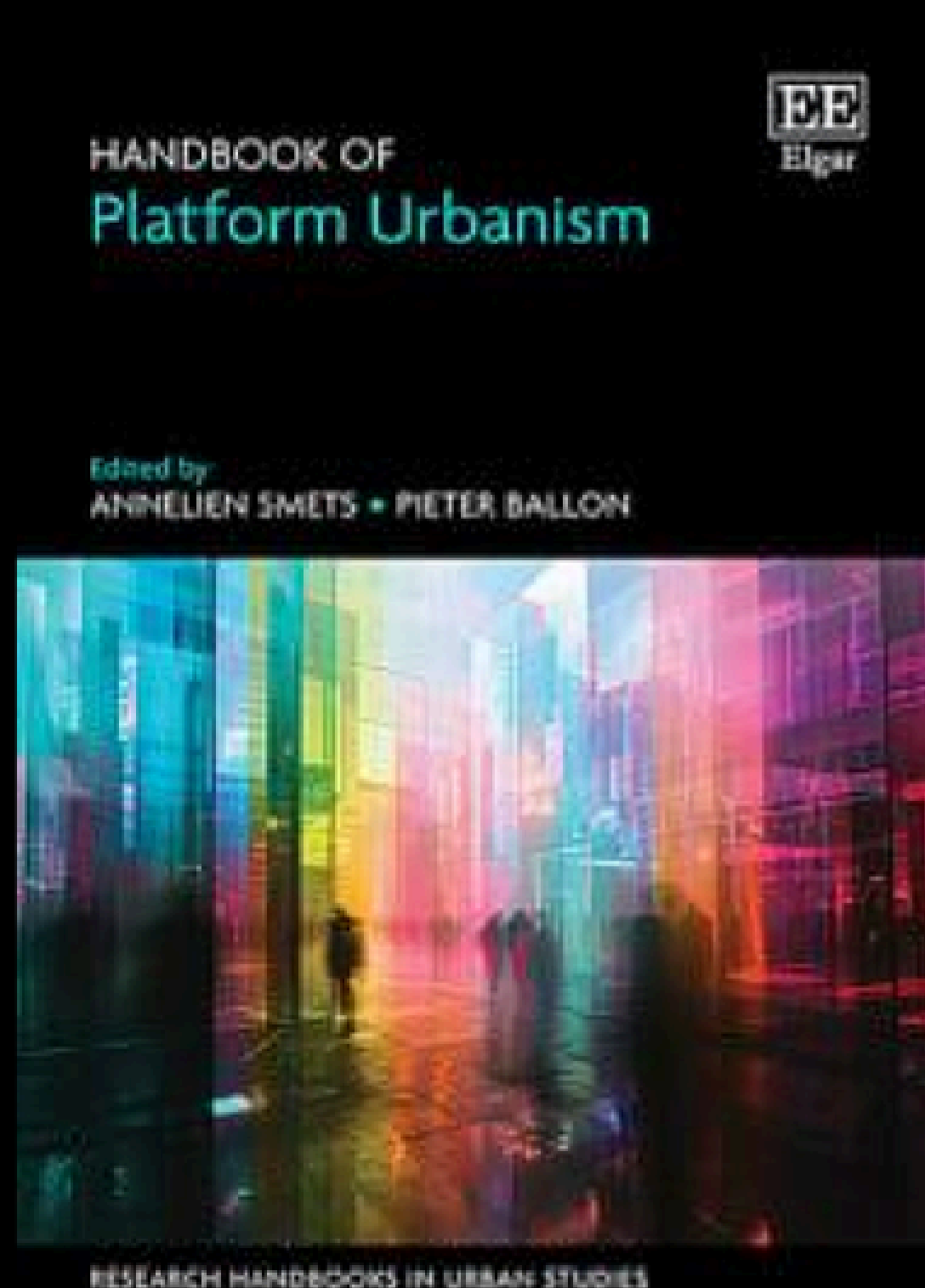
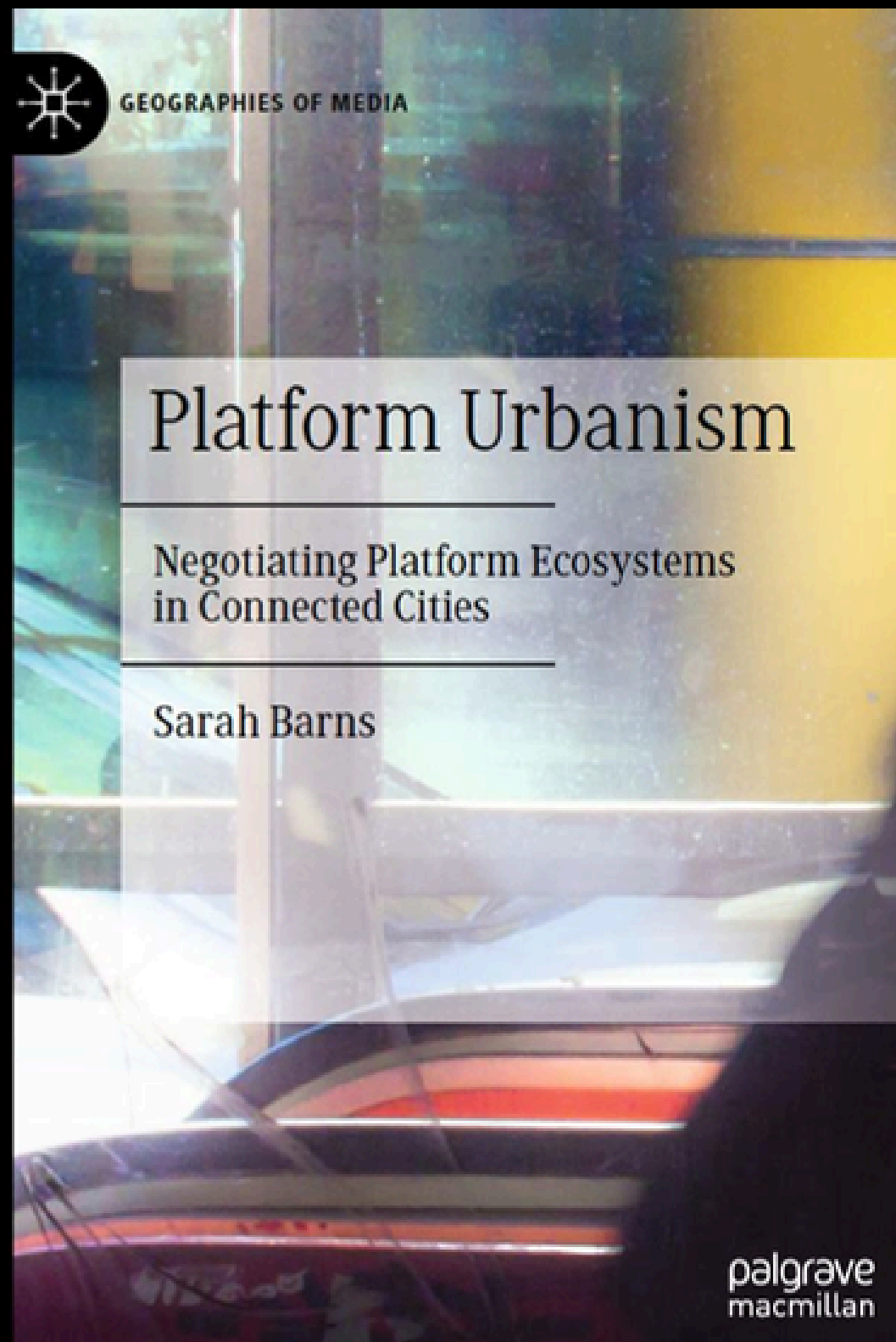


From platform urbanism to AI sovereignty

- > From 'the shadows' of diffuse platform power to a more deliberate and assertive politics of AI sovereignty
- > **To Civic AI?** Counter-actions and epistemic traumas



In the cybernetic shades: Superorganism by ESEM (2022)



Platform Urbanism

Diffuse practices of reterritorialisation through platform interfaces as (digital) intermediaries of urban action

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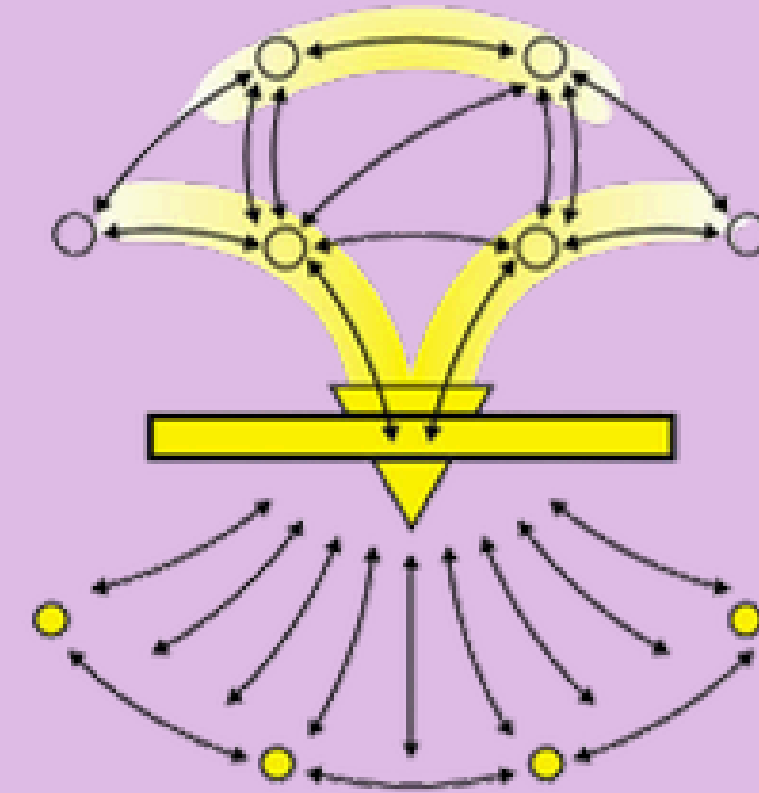
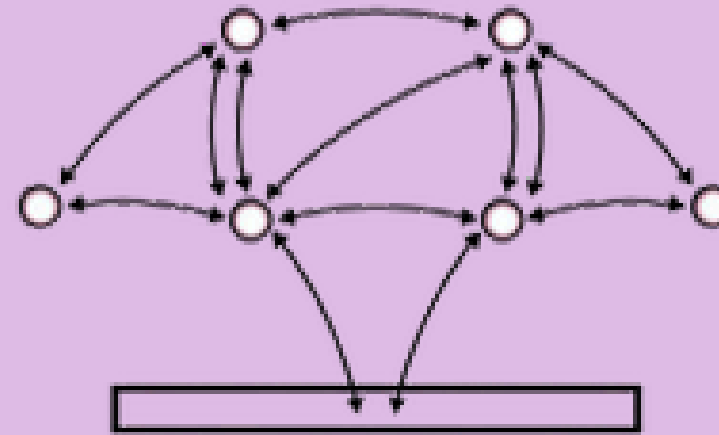
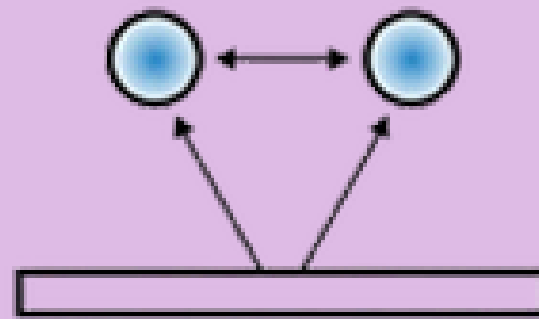
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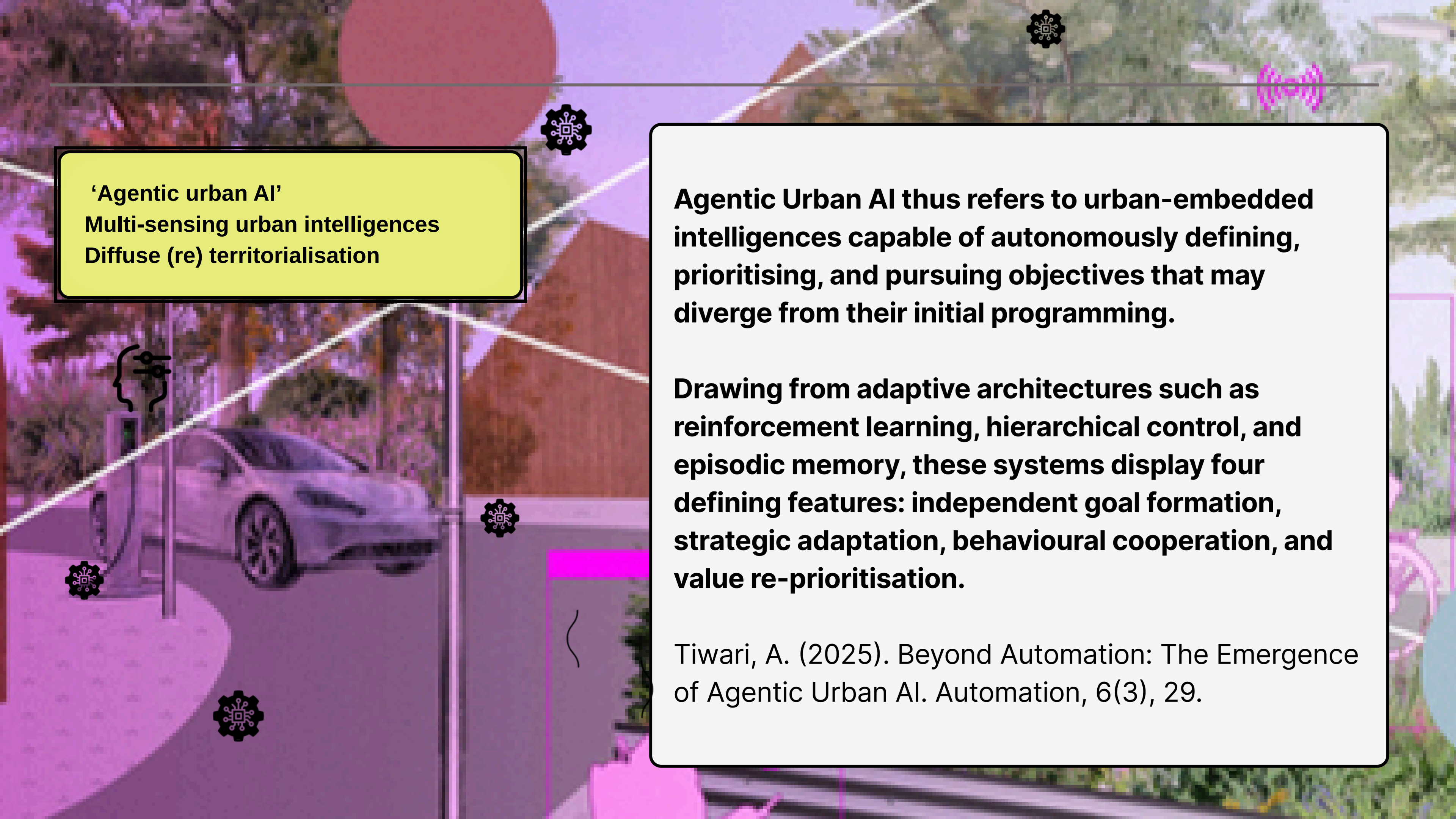
Above the surface: network effects & innovation services



Diffuse power of platforms. From Platform Urbanism (2019)

Below the surface (of digital intermediation) is the accumulation of data intelligence as governing protocols

Platform urbanism has reconstituted the urban terrain as training grounds for urban agentic intelligence

The background image shows a city street scene with a car parked on the side. Overlaid on this image are several semi-transparent shapes and icons. A large red circle is in the upper left. A yellow box with a black border is on the left, containing text. A large white box with a black border is on the right, containing two paragraphs of text and a citation. There are several black gear-like icons with circuit patterns inside, and a pink bow-like icon. Thin white lines connect some of the icons and shapes.

‘Agentic urban AI’
Multi-sensing urban intelligences
Diffuse (re) territorialisation

Agentic Urban AI thus refers to urban-embedded intelligences capable of autonomously defining, prioritising, and pursuing objectives that may diverge from their initial programming.

Drawing from adaptive architectures such as reinforcement learning, hierarchical control, and episodic memory, these systems display four defining features: independent goal formation, strategic adaptation, behavioural cooperation, and value re-prioritisation.

Tiwari, A. (2025). Beyond Automation: The Emergence of Agentic Urban AI. *Automation*, 6(3), 29.

Urban AI
Emerging urban condition?

An emerging urban condition where AI-driven systems are taking on roles traditionally managed by humans in governance, planning, and service provision

Cugurullo, F., Caprotti, F., Cook, M., Karvonen, A., McGuirk, P., & Marvin, S. (Eds.). (2023). Artificial Intelligence and the City: Urbanistic Perspectives on AI (1st ed.). Routledge.

...

The continuities between existing forms of territorial remediation should not be 'forgotten' in the rush to diagnose what is new in this age of urban AI.

Urban AI
[Re]claiming the territory

There is “a prevailing tendency to conceive of AI primarily as an abstract, disembodied form of computational power, with materiality treated as a somewhat secondary medium enabling its execution, rather than as constitutive of its operations.”

Iapaolo, F., & Lynch, C. R. (2025). Rethinking urban AI: a critical posthumanist perspective on intelligence, autonomy, and agency. *Urban Geography*, 1–23.

...

Claiming the territory 'AI Sovereignty'

Territorial claims are made overt -
fused with national sovereignty
agenda

"AI sovereignty is a nation's or organisation's ability to have control over its artificial intelligence technology, including the infrastructure, data, and talent used to create and operate it. This concept is driven by the need for autonomy, national security, economic competitiveness, and alignment with local laws, ethics, and values, especially concerning sensitive data"

National AI Sovereignty Discussion Paper,
Technology Policy Design Institute
Sept 2025

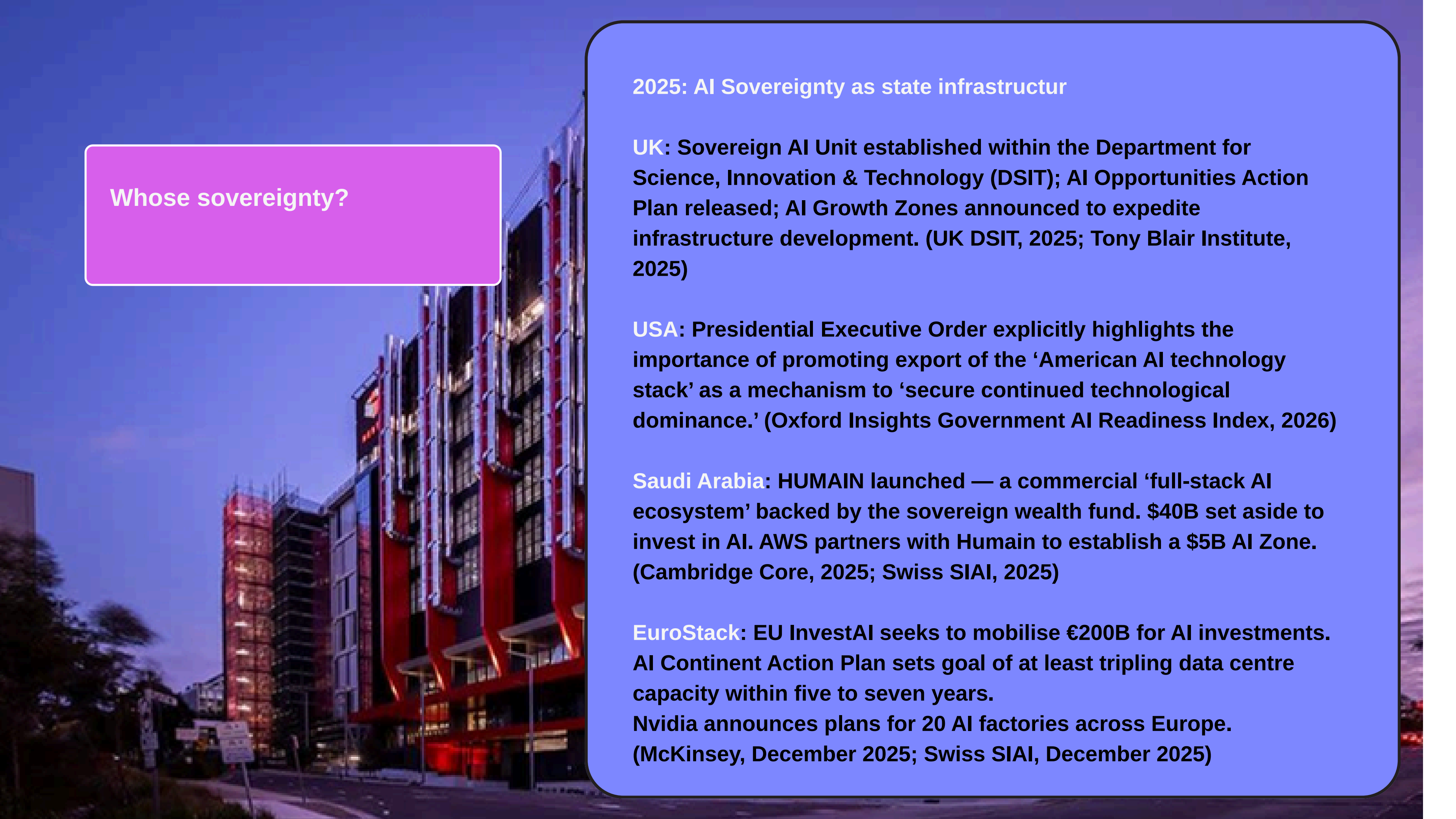
Claiming the territory 'AI Sovereignty'

Every country needs to own the production of their own intelligence... You own your own data. You, therefore, must take that data, refine that data, and own your own national intelligence.' (Nvidia Blog, February 2024)

Jensen Huang (Nvidia) has been pitching 'sovereign AI' since at least 2023, conscious that nation-states are the next deep pockets to target after the hyperscalers." (Cambridge Core, Natural Language Processing, 2025)

**Nation states as
subjects of AI
sovereignty**





Whose sovereignty?

2025: AI Sovereignty as state infrastructure

UK: Sovereign AI Unit established within the Department for Science, Innovation & Technology (DSIT); AI Opportunities Action Plan released; AI Growth Zones announced to expedite infrastructure development. (UK DSIT, 2025; Tony Blair Institute, 2025)

USA: Presidential Executive Order explicitly highlights the importance of promoting export of the 'American AI technology stack' as a mechanism to 'secure continued technological dominance.' (Oxford Insights Government AI Readiness Index, 2026)

Saudi Arabia: HUMAIN launched — a commercial 'full-stack AI ecosystem' backed by the sovereign wealth fund. \$40B set aside to invest in AI. AWS partners with Humain to establish a \$5B AI Zone. (Cambridge Core, 2025; Swiss SIAI, 2025)

EuroStack: EU InvestAI seeks to mobilise €200B for AI investments. AI Continent Action Plan sets goal of at least tripling data centre capacity within five to seven years.

Nvidia announces plans for 20 AI factories across Europe. (McKinsey, December 2025; Swiss SIAI, December 2025)

An architectural rendering of a proposed data centre at dusk. The building is a multi-story structure with a modern design, featuring large glass windows and prominent red vertical accents. A construction crane is visible on the roof. In the foreground, there is a grey wall with a gate and a sign that reads 'NEXTDC'. The sky is a deep blue with some clouds.

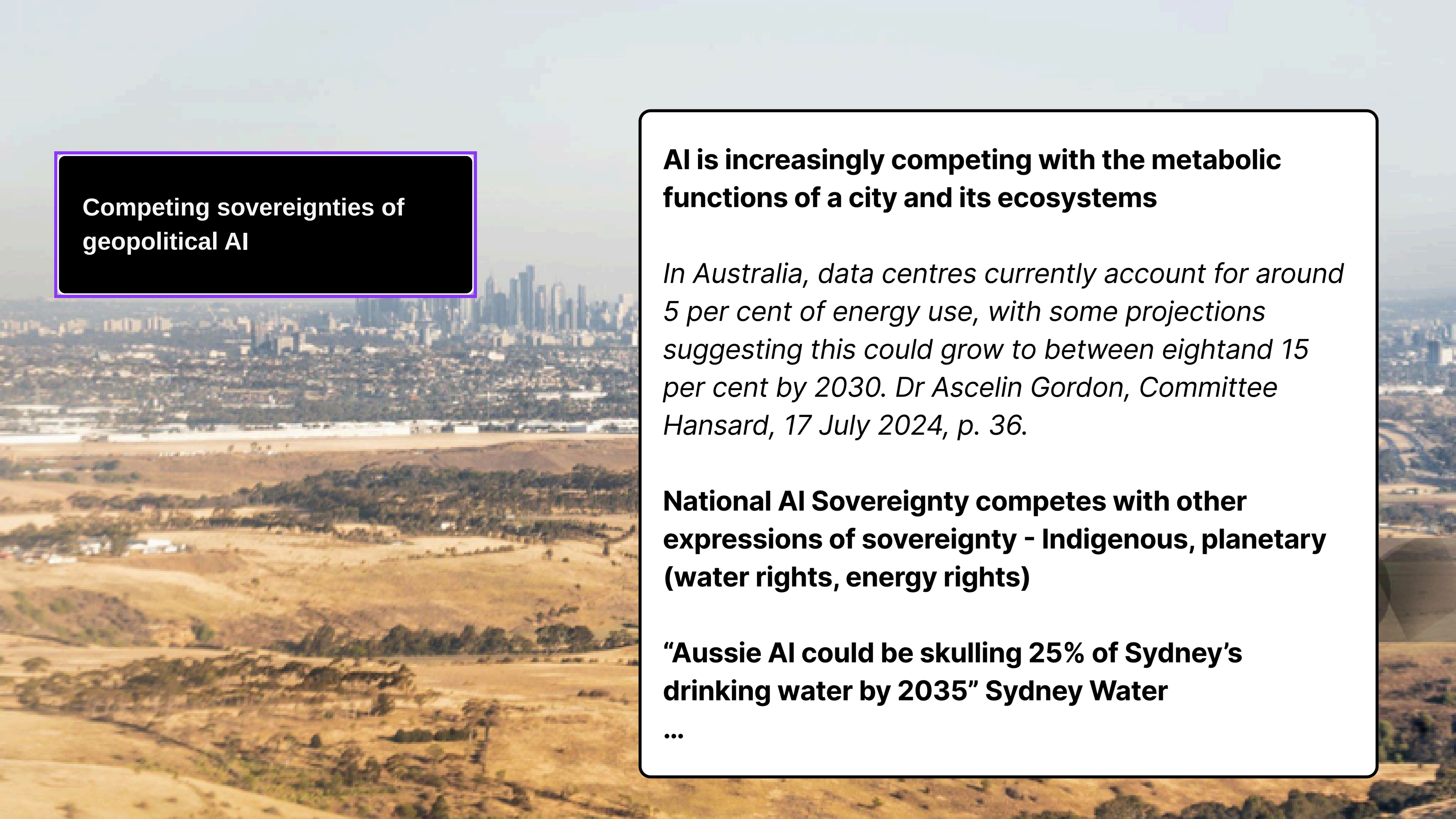
Whose sovereignty?
Acknowledging *the trouble*

NSW Premier Chris Minns, Oct 2025

“ Global investment in these centres will exceed \$6 trillion, by the end of the decade”

“It’s one of the reasons why we’ve introduced the idea to sweep through the planning system, and to take those billion-dollar projects on, head on, and guide them through the planning system in our state.”

Image: Proposed Data Centre, Gore Hill (Next DC)...

An aerial photograph showing a city skyline in the distance, with a dry, hilly landscape in the foreground. The city skyline is visible through a hazy atmosphere, and the foreground consists of rolling hills with sparse vegetation.

Competing sovereignties of geopolitical AI

AI is increasingly competing with the metabolic functions of a city and its ecosystems

In Australia, data centres currently account for around 5 per cent of energy use, with some projections suggesting this could grow to between eight and 15 per cent by 2030. Dr Ascelin Gordon, Committee Hansard, 17 July 2024, p. 36.

National AI Sovereignty competes with other expressions of sovereignty - Indigenous, planetary (water rights, energy rights)

“Aussie AI could be skulking 25% of Sydney’s drinking water by 2035” Sydney Water

...

An aerial photograph showing a city skyline in the distance, with a mix of urban development and dry, hilly terrain in the foreground. The sky is clear and blue.

Competing sovereignties of geopolitical AI

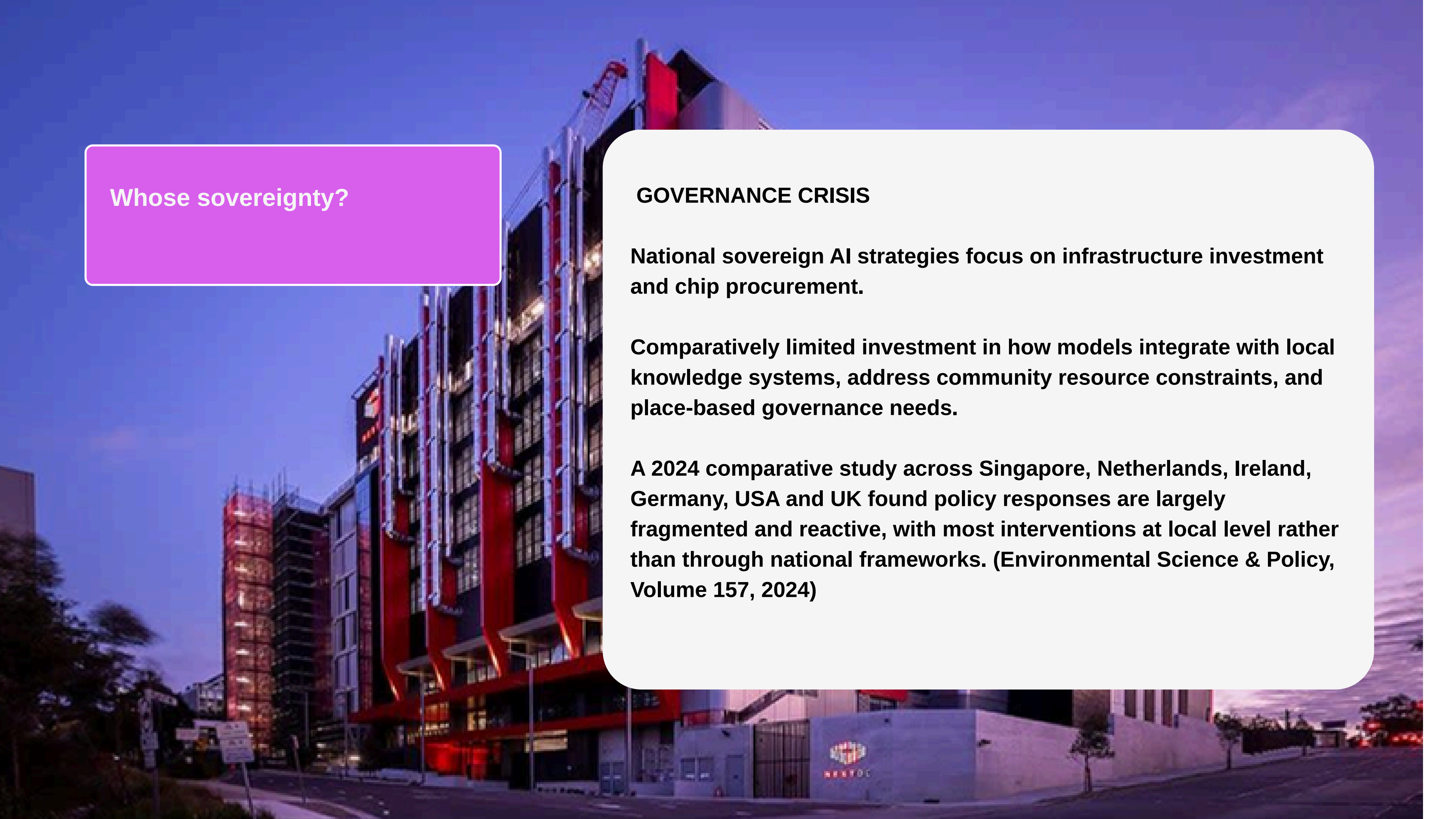
GOVERNANCE CRISIS

National sovereign AI strategies focus on infrastructure investment and chip procurement.

- Limited investment in how models integrate with local knowledge systems, address community resource constraints, and place-based governance needs.

Competing territorial sovereignties are emerging:

- Planetary sovereignty: Who will insure the data centres?
- Indigenous data sovereignty: Whose data
- Place-based sovereignties: 'We live here'
- Moral rights: 'You used my data/IP' in your training model



Whose sovereignty?

GOVERNANCE CRISIS

National sovereign AI strategies focus on infrastructure investment and chip procurement.

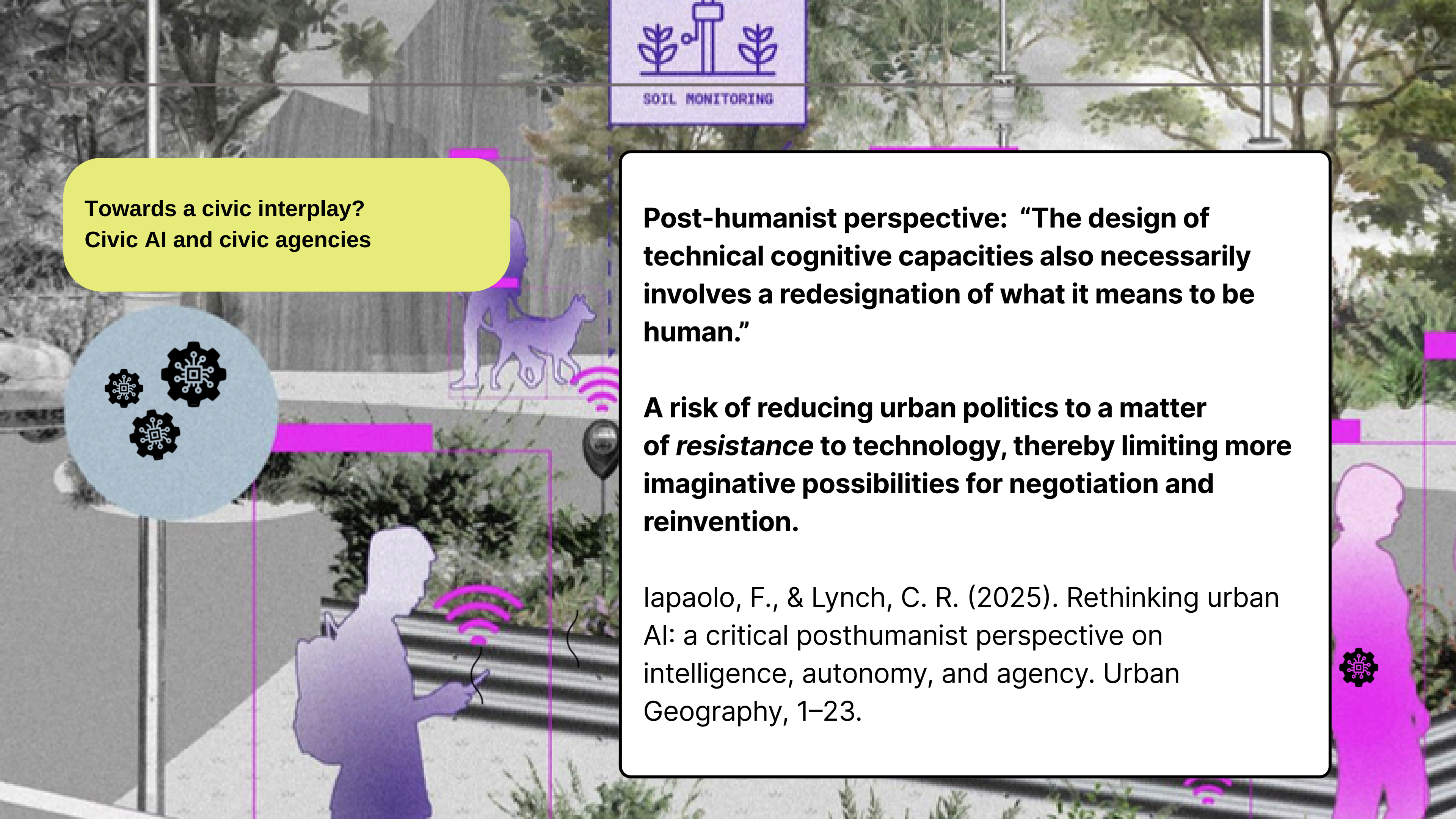
Comparatively limited investment in how models integrate with local knowledge systems, address community resource constraints, and place-based governance needs.

A 2024 comparative study across Singapore, Netherlands, Ireland, Germany, USA and UK found policy responses are largely fragmented and reactive, with most interventions at local level rather than through national frameworks. (Environmental Science & Policy, Volume 157, 2024)

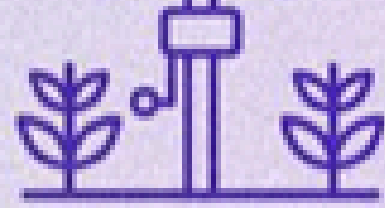
While AI sovereignty is being debated as a technology issue, urban digital scholars like myself understand the critical role that cities play as laboratories for digital transformation. They are sites where citizens, business, communities and governments build successful digital ecosystems and services.

Many AI sovereignty efforts are focused at the urban scale - in cities like Amsterdam, Barcelona, London and Singapore - which lead in advancing citizen-focused tools for new AI literacies.

A comparative urban AI research agenda is needed to examine the role of geopolitics in shaping localised **urban-technology transitions as core testing grounds for AI sovereignty.**



Towards a civic interplay?
Civic AI and civic agencies



SOIL MONITORING

Post-humanist perspective: “The design of technical cognitive capacities also necessarily involves a redesignation of what it means to be human.”

A risk of reducing urban politics to a matter of *resistance* to technology, thereby limiting more imaginative possibilities for negotiation and reinvention.

Iapaolo, F., & Lynch, C. R. (2025). Rethinking urban AI: a critical posthumanist perspective on intelligence, autonomy, and agency. *Urban Geography*, 1–23.

**Towards a civic interplay?
Civic AI and civic agencies**

Training grounds for micro-political imaginaries

Our interconnected lives operate as training grounds for a new kind of planetary intelligence.

The question is not whether we'll have planetary-scale, sovereign AI systems—we already do—but what kind of citizens we'll be in relation to them?

Now is the time to articulate the multiple and competing sovereignties that structure our digital urban lives

**Towards a civic interplay?
Civic AI and civic agencies**

- How do citizens, governments, community and advocacy groups, businesses actually collaborate to develop the literacies and capabilities needed for an era of AI and data sovereignty?
- What conditions restrict or support decision-making autonomy over how technologies are deployed in urban settings? Are they legal, governance, policy, social, financial, ecological, cultural conditions?
- How are the supporting conditions made sustainable over time, through toolkits, diagnostic frameworks, policies and procurement frameworks?

CIVIC INTERPLAY

MICRO POLITICS FOR A MACRO AGE

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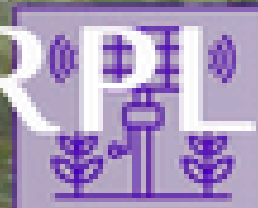
SARAH BARNES



ROAD SENSORS



AIR QUALITY SENSORS



TREE MONITORING